

# Quantum Computing Fundamentals

From kets to key distribution

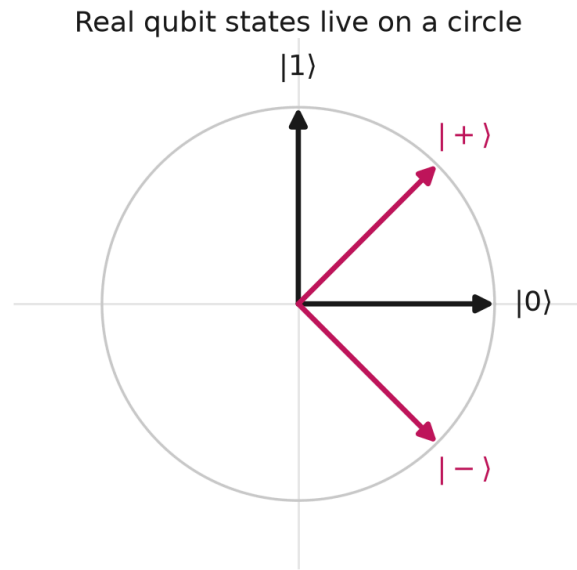
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Based on *Quantum Computing for Everyone* (Bernhardt), implemented in Qiskit

## Everything here is 2D, and real

A qubit is a vector of length 1. With **real** amplitudes, every state sits on the unit circle. **Kets** are column vectors; **bras** are their transpose.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \langle 0| = \begin{bmatrix} 1 & 0 \end{bmatrix}$$



Complex amplitudes would fill a sphere instead. Not needed until the quantum Fourier transform.

## A qubit, and why the squares sum to 1

$$|v\rangle = c_0|0\rangle + c_1|1\rangle \quad c_0^2 + c_1^2 = 1$$

$c_0$  and  $c_1$  are **probability amplitudes**. Measuring gives

$$P(0) = c_0^2 \quad P(1) = c_1^2$$

so the normalisation condition is just "the probabilities add to 1".

**Standard basis** —  $\{|0\rangle, |1\rangle\}$  — is orthonormal:

$$\langle 0|0\rangle = 1 \quad \langle 1|1\rangle = 1 \quad \langle 0|1\rangle = 0$$

A bra times a ket gives a number: the overlap between two states.

## Two qubits: the tensor product

Alice has  $|v\rangle = c_0|a_0\rangle + c_1|a_1\rangle$ , Bob has  $|w\rangle = d_0|b_0\rangle + d_1|b_1\rangle$ .

$$|v\rangle \otimes |w\rangle = c_0d_0|a_0b_0\rangle + c_0d_1|a_0b_1\rangle + c_1d_0|a_1b_0\rangle + c_1d_1|a_1b_1\rangle$$

Name the four amplitudes:

$$r = c_0d_0 \quad s = c_0d_1 \quad t = c_1d_0 \quad u = c_1d_1$$

It is still normalised, because

$$\underbrace{(c_0^2 + c_1^2)}_1 \underbrace{(d_0^2 + d_1^2)}_1 = r^2 + s^2 + t^2 + u^2 = 1$$

## The entanglement test

If the state really came from a tensor product, then

$$ru = c_0d_0 \cdot c_1d_1 = c_0c_1d_0d_1 \quad st = c_0d_1 \cdot c_1d_0 = c_0c_1d_0d_1$$

They must be equal. So:

|                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} ru = st &\Rightarrow \text{separable} \\ ru \neq st &\Rightarrow \text{entangled} \end{aligned}$ |
|-------------------------------------------------------------------------------------------------------------------|

Separable means each qubit has a state of its own. Entangled means only the **pair** has a state. Two worked examples follow — one of each.

Equivalently:  $ru - st$  is the determinant of  $\begin{bmatrix} r & s \\ t & u \end{bmatrix}$ , so separable = determinant zero.

## Example A — not entangled

Alice  $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ , Bob  $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ . Tensor them:

$$r = \frac{1}{2\sqrt{2}} \quad s = \frac{\sqrt{3}}{2\sqrt{2}} \quad t = \frac{1}{2\sqrt{2}} \quad u = \frac{\sqrt{3}}{2\sqrt{2}}$$

**Test:**  $ru = \frac{\sqrt{3}}{8}$  and  $st = \frac{\sqrt{3}}{8}$ . Equal  $\rightarrow$  separable.

**Group by Alice's kets:**

$$|a_0\rangle \left( \frac{1}{2\sqrt{2}} |b_0\rangle + \frac{\sqrt{3}}{2\sqrt{2}} |b_1\rangle \right) + |a_1\rangle \left( \frac{1}{2\sqrt{2}} |b_0\rangle + \frac{\sqrt{3}}{2\sqrt{2}} |b_1\rangle \right)$$

**Normalise each bracket** — both have norm  $\sqrt{\frac{1}{8} + \frac{3}{8}} = \frac{1}{\sqrt{2}}$  — and Bob's part is a **common factor**:

$$= \left( \frac{1}{\sqrt{2}} |a_0\rangle + \frac{1}{\sqrt{2}} |a_1\rangle \right) \left( \frac{1}{2} |b_0\rangle + \frac{\sqrt{3}}{2} |b_1\rangle \right)$$

Each qubit has a state of its own. Now the same four steps on one that fails.

## Example B — entangled

$$r = \frac{1}{2} \quad s = \frac{1}{2} \quad t = \frac{1}{\sqrt{2}} \quad u = 0$$

**Test:**  $ru = 0$  but  $st = \frac{1}{2\sqrt{2}}$ . Different  $\rightarrow$  entangled.

**Group by Alice's kets and normalise the same way:**

$$\frac{1}{\sqrt{2}}|a_0\rangle \left( \frac{1}{\sqrt{2}}|b_0\rangle + \frac{1}{\sqrt{2}}|b_1\rangle \right) + \frac{1}{\sqrt{2}}|a_1\rangle|b_0\rangle$$

Bob's two brackets are now **different**, so nothing factors out. Neither qubit has a state of its own.

**What that means:** Alice measures  $|a_0\rangle$  or  $|a_1\rangle$  with equal odds — but her result decides Bob's. Get  $|a_1\rangle$  and Bob is  $|b_0\rangle$  with certainty.

# Gates

So far: how to **describe** a two-qubit state, and test whether it is entangled.

Now: how to **change** one — and how entanglement actually gets made.



## $I$ and $Z$ — leave the magnitudes alone

All of these act on  $a_0|0\rangle + a_1|1\rangle$  in the standard basis.

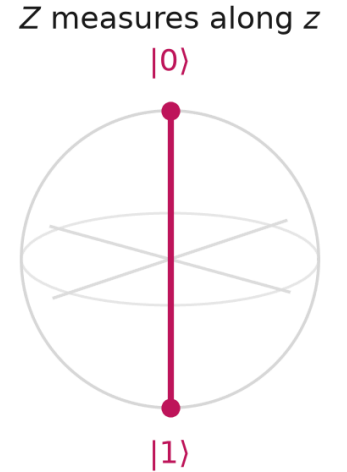
$I$  is the **identity** — it does nothing:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I(a_0|0\rangle + a_1|1\rangle) = a_0|0\rangle + a_1|1\rangle$$

$Z$  leaves both **magnitudes** unchanged but flips the sign of  $a_1$  — the **relative phase**:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad Z(a_0|0\rangle + a_1|1\rangle) = a_0|0\rangle - a_1|1\rangle$$

Probabilities are unaffected, since  $|-a_1|^2 = |a_1|^2$ . The phase only becomes visible once the qubit **interferes** with another — for instance after a Hadamard.



## $X$ and $Y$ — the two NOTs

Both exchange  $|0\rangle$  and  $|1\rangle$ .

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$X(a_0|0\rangle + a_1|1\rangle) = a_1|0\rangle + a_0|1\rangle \quad Y(a_0|0\rangle + a_1|1\rangle) = a_1|0\rangle - a_0|1\rangle$$

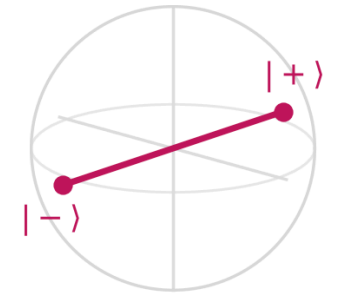
So  $X$  is a pure swap;  $Y$  swaps **and** changes the relative phase.

The Pauli  $Y$  is usually  $-i$  times the matrix above, giving  $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ . The real version is used here because these slides avoid complex numbers — but only the complex one squares to  $I$ . Qiskit's `qc.y()` applies the complex one.

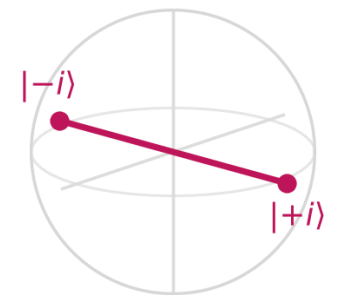
Three spatial axes, so **three** Paulis — not one  $3 \times 3$  matrix but three separate  $2 \times 2$  ones.

The size comes from the number of **outcomes**: measuring along any axis gives two.

$X$  measures along  $x$



$Y$  measures along  $y$

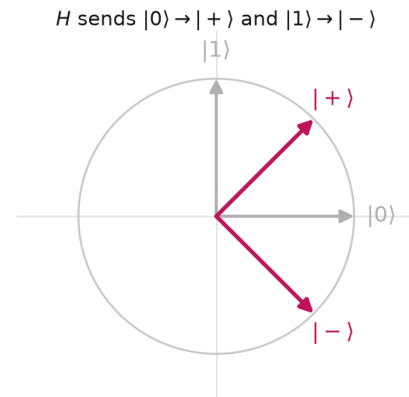
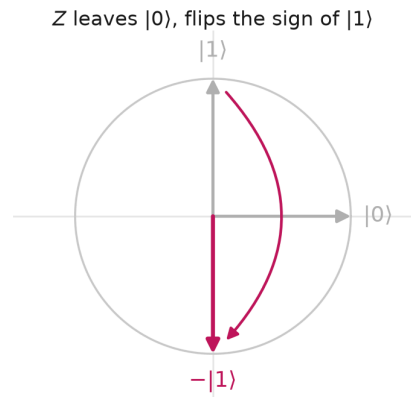


## Hadamard — into superposition

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$



In a diagram, a one-qubit gate is a square with its letter in it:



## Why Hadamard, specifically?

The  $\frac{1}{\sqrt{2}}$  is not a choice. Without it  $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} |0\rangle$  has  $\text{length}^2 = 2$ , so the probabilities would sum to 2. Every gate must preserve length.

### Plenty of gates make superposition:

|                        | $P(0)$ | $P(1)$ |
|------------------------|--------|--------|
| $R_y(\pi/3) 0\rangle$  | 0.75   | 0.25   |
| $R_y(\pi/2) 0\rangle$  | 0.50   | 0.50   |
| $R_y(2\pi/3) 0\rangle$ | 0.25   | 0.75   |

$R_y(\pi/2)$  gives the **same** equal superposition as  $H$  does from  $|0\rangle$ .

### But only $H$ undoes itself:

$$H^2 = I \quad R_y(\pi/2)^2 = R_y(\pi) \neq I$$

They agree on  $|0\rangle$  and differ on  $|1\rangle$  — that difference *is* the involution.

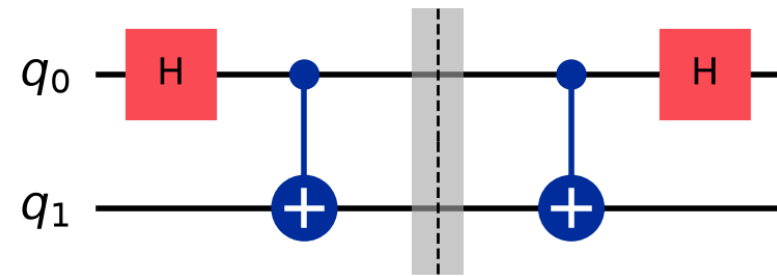
$H$  is a **basis change** between  $Z$  and  $X$ , and swapping bases twice returns you home.

Every protocol here leans on that: the Bell circuit, Bob's decoder, BB84's  $X$ -basis measurement.

# Involutions — gates that undo themselves

Applying one twice returns the input:  $U^2 = I$

| Gate    | Qubits | What it does                                |
|---------|--------|---------------------------------------------|
| $X$     | 1      | swaps $ 0\rangle \leftrightarrow  1\rangle$ |
| $Y$     | 1      | swaps, with phases                          |
| $Z$     | 1      | sign flip on $ 1\rangle$                    |
| $H$     | 1      | basis change                                |
| CNOT    | 2      | flip if control is 1                        |
| SWAP    | 2      | exchange the two                            |
| Toffoli | 3      | flip if both controls 1                     |
| Fredkin | 3      | swap if control is 1                        |

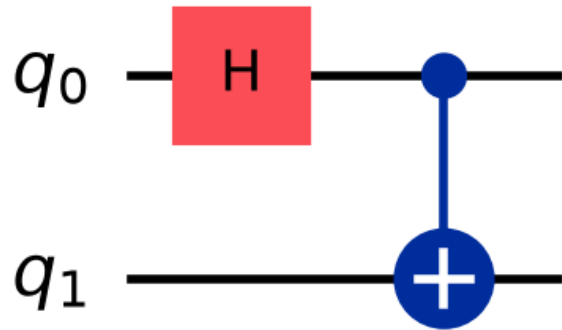


At the barrier the state is  $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$ .  
Run the same two gates again and you are back at  $|00\rangle$ .

That "un-preparing" step is exactly what Bob does in superdense coding, and Alice in teleportation.

## CNOT, and making entanglement

A single-qubit gate can **never** entangle. You need a two-qubit gate.

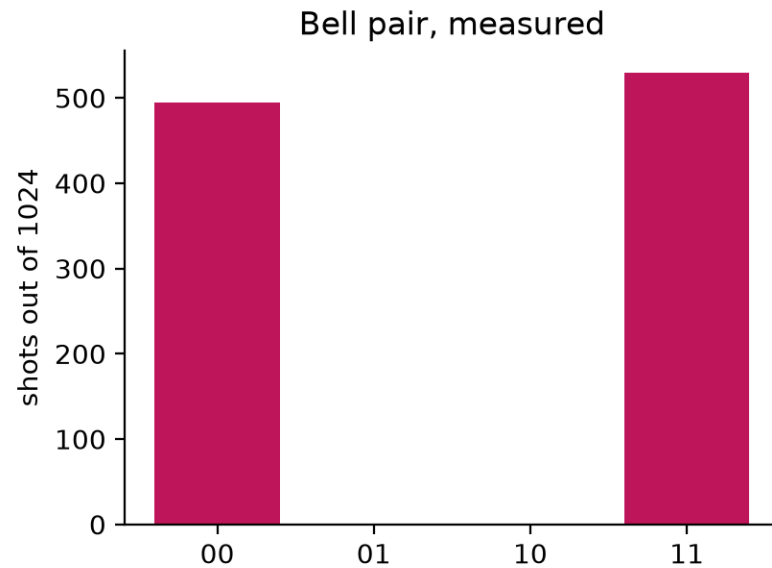


```
qc = QuantumCircuit(2)
qc.h(0)      # superposition
qc.cx(0, 1)  # entangle
```

$$|00\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Test it:  $r = u = \frac{1}{\sqrt{2}}$ ,  $s = t = 0$ , so  $ru = \frac{1}{2} \neq 0 = st$ . **Entangled.**

## And it shows up in the measurements



Only `00` and `11` . Never `01` or `10` .

The two qubits always agree — but neither had a value before measurement.

This **Bell pair** is the resource every protocol below is built on.

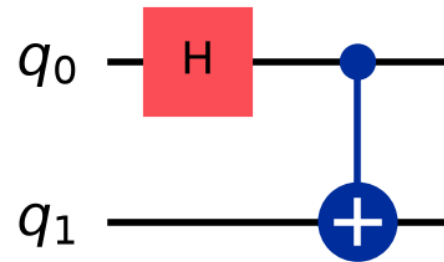
# Protocols

We can now **make** a Bell pair. What is it good for?

Two protocols that trade one resource against the other:

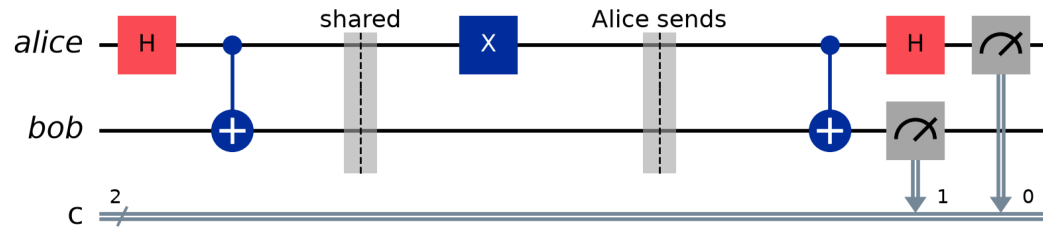
**Superdense coding** — 2 classical bits for 1 qubit

**Teleportation** — 1 qubit for 2 classical bits



# Superdense coding

Send **2 classical bits** by physically transmitting **1 qubit**.



**Alice owns q0, Bob owns q1** — they share  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , one each.

Alice applies one gate to **her qubit only**, sends it to Bob, and he — now holding both — undoes the Bell circuit.

| Bob finally reads | 00  | 01  | 10  | 11   |
|-------------------|-----|-----|-----|------|
| so Alice applies  | $I$ | $X$ | $Z$ | $ZX$ |

The top row is the **message Bob reads after decoding** — not the state in flight.

## Superdense coding — why it works

$$|00\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice applies  $X$  to send `01`:

$$\xrightarrow{X} \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$$

Bob reverses the Bell circuit:

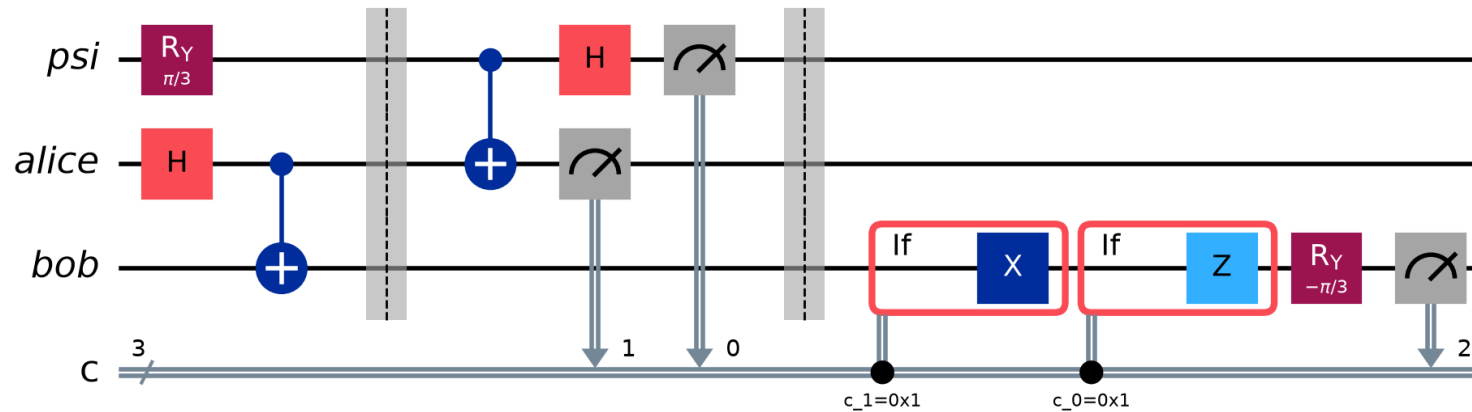
$$\xrightarrow{CNOT} \frac{1}{\sqrt{2}}(|11\rangle + |01\rangle) \xrightarrow{H} |01\rangle \quad \text{— the two bits he reads}$$

```
qc.h(0); qc.cx(0, 1)    # Bell pair, shared in advance
qc.x(0)                 # Alice encodes "01" on her qubit alone
qc.cx(0, 1); qc.h(0)    # Bob decodes
```

The four gates map the pair onto four distinguishable Bell states. Entanglement is what lets one qubit carry two bits.

# Quantum teleportation

Move **1 qubit's state** using **2 classical bits**. The exact opposite trade.



**Alice owns *psi* and *alice* ; Bob owns *bob* .** She entangles her state with her half of the pair, measures both of hers, and sends two classical bits — nothing quantum crosses. Bob corrects his qubit.

| Bob receives | 00  | 01  | 10  | 11  |
|--------------|-----|-----|-----|-----|
| Bob applies  | $I$ | $X$ | $Z$ | $Y$ |

## Teleportation — the catch

The state is **not copied** — no-cloning forbids that. Alice's measurement destroys her copy at the moment Bob's appears.

Nothing travels faster than light either: without the 2 classical bits, Bob's qubit is useless. Each of his four possible states is equally likely.

```
qc.ry(theta, psi)           # the state to send
qc.h(a); qc.cx(a, b)        # shared Bell pair
qc.cx(psi, a); qc.h(psi)    # Alice: reverse Bell circuit
qc.measure(psi, c[0]); qc.measure(a, c[1])
with qc.if_test((c[1], 1)): qc.x(b)    # Bob corrects
with qc.if_test((c[0], 1)): qc.z(b)
```

Verified in `teleportation.py`:  $|0\rangle$ ,  $|1\rangle$ ,  $|+\rangle$  and  $R_y(\pi/3)|0\rangle$  all arrive in 100% of shots.

# Key distribution

Those two moved information between people who **trust each other**.

Now: two protocols where somebody is **listening** — and the physics says so.

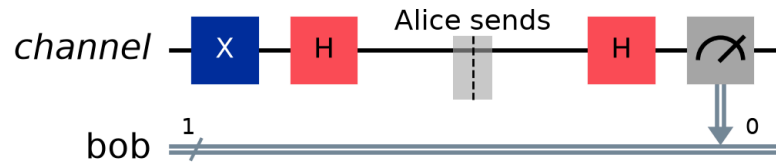
**BB84** — two conjugate bases, no entanglement at all

**E91** — entanglement, and a Bell inequality

## BB84 — detecting an eavesdropper

Not encryption. It produces a shared random key **and tells you if anyone looked**.

$Z = \{|0\rangle, |1\rangle\}$  and  $X = \{|+\rangle, |-\rangle\}$  are **conjugate** — certain in one, a coin flip in the other.



**One qubit, changing hands.** `channel` is Alice's before the barrier, Bob's after.

- `X` — Alice's **bit**: present 1, absent 0
- first `H` — her **basis**: present  $X$ , absent  $Z$
- second `H` — **Bob's** basis, picked blind

**The order is the security:**

1. Alice sends the qubit
2. Bob measures **immediately**, guessing a basis
3. *Then* they announce bases — never bits
4. Discard the mismatches

Bases match  $\rightarrow$  the two  $H$ s cancel,  $H^2 = I \rightarrow$  Bob reads her bit exactly. Bases differ  $\rightarrow$  one lone  $H \rightarrow$  a coin flip.

## BB84 — eight rounds, start to finish

Alice flips **two** coins (bit, basis). Bob flips **one** (basis), blind.

| # | A bit | A base | sends       | B base | B reads  | match? |                         |
|---|-------|--------|-------------|--------|----------|--------|-------------------------|
| 0 | 1     | $X$    | $ -\rangle$ | $X$    | <b>1</b> | yes    | <b>key</b>              |
| 1 | 0     | $Z$    | $ 0\rangle$ | $X$    | 0        | no     | discard — right by luck |
| 2 | 0     | $Z$    | $ 0\rangle$ | $X$    | 1        | no     | discard — <b>wrong</b>  |
| 3 | 1     | $Z$    | $ 1\rangle$ | $Z$    | <b>1</b> | yes    | <b>key</b>              |
| 4 | 0     | $X$    | $ +\rangle$ | $X$    | <b>0</b> | yes    | <b>key</b>              |
| 5 | 0     | $Z$    | $ 0\rangle$ | $Z$    | <b>0</b> | yes    | <b>key</b>              |
| 6 | 0     | $Z$    | $ 0\rangle$ | $Z$    | <b>0</b> | yes    | <b>key</b>              |
| 7 | 0     | $Z$    | $ 0\rangle$ | $X$    | 0        | no     | discard                 |

They announce their **bases** publicly — never the bits — and drop the mismatches.

Alice keeps **1 1 0 0 0**, Bob keeps **1 1 0 0 0** : a secret neither of them sent.

Rounds 1 and 2 are the point — same choices, different outcomes. On a mismatch Bob still gets a bit, just a worthless one, and nobody knows which until the bases are announced.

## BB84 — what actually becomes key

A real run of `bb84.py --rounds 10` :

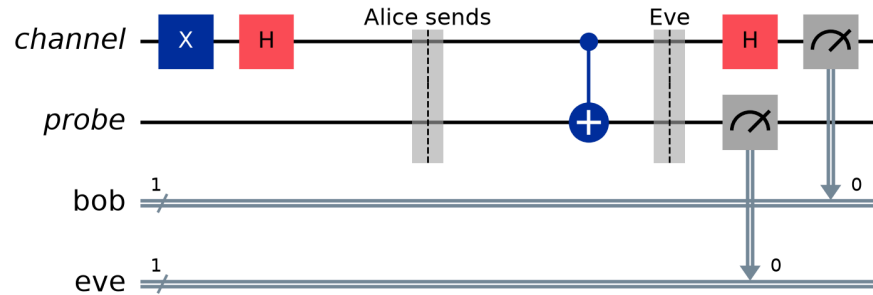
```
transmitted          10
  bases disagreed      3   dropped at sifting
  bases agreed        7   = the sifted bits
    sacrificed        4   announced openly, so spent either way
    left for the key   3   <- the key
```

**Two losses.** Half the rounds die at sifting, because the bases disagreed. More are sacrificed to test for Eve — once announced they are public, so they are spent whatever they show. Roughly **a quarter** of transmitted qubits become key.

Each compared bit is an independent  $\frac{3}{4}$  chance Eve slips past, so detection needs volume:

| bits compared          | 4   | 50                 | 126                 |
|------------------------|-----|--------------------|---------------------|
| Eve escapes undetected | 32% | $6 \times 10^{-7}$ | $2 \times 10^{-16}$ |

## BB84 — what Eve costs



Eve cannot copy the qubit (**no-cloning**), so she must measure — and guess a basis.

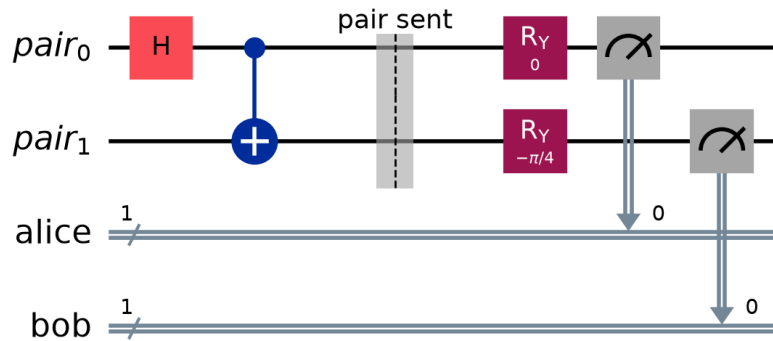
She is wrong half the time, and then Bob errs half of *those*:

$$\text{QBER} = \frac{1}{2} \times \frac{1}{2} = 25\%$$

She does **not** change the sifting rate — only the error rate on the bits that survive:  $0\% \rightarrow 25\%$ . Alice and Bob compare some openly, and **abort above 11%**.

Her gain: 75% of the key — useless, because the key is discarded.

## E91 — the same job, via Bell's theorem



A source emits entangled pairs. Each side measures along a randomly chosen axis.

$$E(\theta_a, \theta_b) = \cos(\theta_a - \theta_b)$$

Combine four settings into the **CHSH** sum  $S$ . Any theory where the bits existed before measurement obeys  $|S| \leq 2$ . Quantum mechanics reaches  $2\sqrt{2} \approx 2.83$ .

Eve has to entangle herself with the pair to learn anything — and **monogamy of entanglement** means that halves every correlation, dropping  $S$  to 1.41.

Measured:  $S = 2.78 \pm 0.07$  clean, 1.34 with Eve. The eavesdropper is visible as a *failed violation*.

## Recap

|                   | <b>sends</b>       | <b>adversary</b> | <b>needs entanglement</b> |
|-------------------|--------------------|------------------|---------------------------|
| Superdense coding | 2 bits via 1 qubit | no               | yes                       |
| Teleportation     | 1 qubit via 2 bits | no               | yes                       |
| BB84              | a secret key       | <b>yes</b>       | no                        |
| E91               | a secret key       | <b>yes</b>       | yes                       |

All four run locally or on IBM hardware with `--ibm` .

[github.com/alessandrogpz/quantum-computing-notes](https://github.com/alessandrogpz/quantum-computing-notes)